SOMAIYA
VIDYAVIHAR UNIVERSITY

23-05-2025(E)

Maximum Marks: 100	Semester: January 2025 –April 2025 Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01 Programme: B. Tech Computer Engineering	Class: SY	Semester:III(SVU 2020)
Name of the Constituent College/School K. J. Somaiya School of Engineering	Name of the department: Computer	
Course Code: 116U01C304	Name of the Course: Object Oriented Programming Methodology	

Instructions: 1)Give neat code snippets as examples
2) All questions are compulsory
3) Assume suitable data wherever necessary

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	What is method overloading in Java? Demonstrate its use with a relevant program and discuss when it is useful.	5
ii)	Differentiate between deep copy and shallow copy in Java with the help of code snippets.	5
iii)	How does encapsulation help in data security in Java? Illustrate with an appropriate example.	5
iv)	Explain the use of static blocks in Java. How are they different from constructors?	5
v)	Write a Java code snippet to convert an array to a List and vice versa.	5
vi)	Discuss the significance of Java's final keyword. Give examples to show its usage with variables, methods, and classes.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Write a Java program that accepts a list of employee names and their joining years and prints the names of employees who have completed more than 5 years in the company as of the current year.	5
ii)	Write a Java program that reads a number from the user and prints whether the number is a **prime palindrome** or not. (A prime palindrome is a number that is both prime and a palindrome.)	5
	OR	
Q2 A	Write a Java program that accepts a list of strings and checks whether any string is a **palindromic permutation** (a permutation that can form a palindrome). Return true if at least one exists, otherwise false.	10

Q 2 B	Solve any **One**	10
i)	Design a class hierarchy to model the following scenario: An institution maintains records of Faculty and Administrative Staff. Faculty have subjects and research areas, while Administrative Staff have departments and working shifts. Use inheritance and polymorphism to model this and write Java class definitions.	10
ii)	Model an app store using object-oriented principles. You should have classes for App, Game (inherits App), and User. Games have a difficulty level and score, while Apps have categories and usage time. Create a class diagram and write relevant Java classes.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	What are Collections in Java? Write a program that stores and retrieves elements of any type from an ArrayList and a Vector.	10
ii)	Write a Java program to read a paragraph and count the number of vowels, consonants, digits, spaces, and special characters.	10
iii)	What is multi-threading in Java? Explain the lifecycle of a thread and demonstrate creating two threads that print odd and even numbers from 1 to 20.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Create a Java program to simulate a university system where multiple students (threads) try to register for the same course concurrently. Implement synchronization to avoid duplicate enrolments.	10
ii)	Write a Java program to manage a student attendance system using ArrayList. Allow operations such as adding student names, marking attendance, and displaying the list of students who were present.	10
iii)	Define and use a custom checked exception called InvalidScoreException, which is thrown when a score entered is less than 0 or more than 100. Use try-catch to handle it and display appropriate messages.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any **four**	**20**
i)	Difference between String, StringBuffer, and StringBuilder with examples.	5
ii)	Role of constructors in inheritance with an example.	5
iii)	Access specifiers in Java	5
iv)	Interfaces vs Abstract classes with at least two examples.	5
v)	Final, finally, and finalize – Explain the difference with examples.	5
vi)	Use of ***this*** keyword in Java.	5